When Is a Photovoltaic Defect Detector Independently Validated?

Dataset Ancestry, Model Selection and Experiment-Indexed Evidence in Imaging-Based AI Research

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Scientific development manuscript — not the completed systematic review or a submission-ready copy.

Abstract

Performance reported for artificial-intelligence (AI) inspection of photovoltaic (PV) modules is conditional on the objects tested, the model-selection procedure and the denominator of each reported measure. Yet an unseen image is not necessarily an unseen physical module or plant. We develop an experiment-indexed claim-support record linking original study and data versions, acquisition parent, derived-image ancestry, partition unit, checkpoint selection, prediction target, statistical unit and evidence tier. Rather than introducing another algorithm leaderboard, this audit distinguishes author-reported validation protocols from independently replayed physical-parent separation. An illustrative reanalysis of a published multi-plant infrared study's 54 printed AUROC cells reconstructs nine source–target tasks under three checkpoint-selection conditions for two classifiers. On three of nine tasks, the model ordering under target-label-selected "Best" differs from both other-plant validation selections; the largest displayed within-method target-best minus validation-plant differences are 11.71 and 19.73 AUROC percentage points. These are descriptive differences between source-reported selection conditions, not causal leakage estimates or new model runs. Complementary source cases distinguish module-grouped thermal classification, group-aware electroluminescence folds and failed cross-dataset score calibration. The broader review corpus retains 21 original database exports and 8,326 provider records linked provisionally to 5,440 identity groups, but final eligibility and PRISMA reporting remain open. The proposed outcome is a reproducible, evidence-bounded account of what each PV imaging experiment does—and does not—support, rather than a pooled model-performance claim.

Keywords: photovoltaics; electroluminescence; infrared thermography; experimental independence; dataset provenance; model selection; reproducibility; methodological review.

1 The research question and originality boundary

Photovoltaic inspection studies report classification accuracy, F1, AUROC, average precision, mAP and segmentation measures under different acquisition regimes. A thermogram of a module, a crop of one cell, multiple video frames of the same module, a synthetically augmented image, and a newly inspected PV plant are not interchangeable experimental units. A numerical performance figure is therefore evidence about a particular *target, source population, partition and selection rule*; it is not itself a demonstration of external transfer.

Several existing reviews have already summarized PV faults, EL and infrared imaging, machine-learning models, datasets and general field-robustness limitations. Ali, Zafar and Kim provide a systematic imaging-focused review of 2021–2024 studies [3]; Meena and Gawre synthesize 95 AI fault-diagnosis studies across modalities [4]; an IEEE Access review discusses diagnostic techniques and AI datasets and evaluation metrics [5]; and Rahman and Srivastava integrate detection, classification and localization across fault families [6]. Pillai and colleagues' publisher-listed review, indexed for a 2027 volume, explicitly distinguishes fault-visibility benchmarking from naive metric comparison [7]. These reviews are substantive prior work. Consequently, broad statements such as "datasets are small," "more external validation is needed," or "methods cannot be directly compared" are *not* claimed as new contributions here.

Earlier structured comparative work directly overlaps the source-data themes: Li and coauthors already tabulated datasets, original/augmented images and some train–test partitions in a 2021 neural-network PV fault-diagnosis review [8]. Nasri and coauthors compared multimodal datasets and fusion methods at study

level [10]; Wang and Xing reviewed nondestructive PV defect AI [9]; and Lu and coauthors proposed a qualitative multi-indicator comparison of fusion methods [11]. These are direct predecessor contributions, not merely generic warnings about external validation. Their full-text coverage of each evidence field is not yet exhaustively adjudicated, so methodological priority cannot be asserted from abstract silence.

The potential additional contribution is an auditable, *result-specific* bridge from source evidence to admissible inference. We ask not only whether a paper describes AI on EL or IR, but which individual result was obtained under which versioned source, original physical acquisition, partition, checkpoint-selection rule and denominator. One report can therefore generate several different experiment/result records while retaining one bibliographic study identity. The proposed record is an audit instrument, not a composite methodological-quality index or a claim of priority over every preceding review. Whether this is a sufficiently distinct systematic-review contribution remains to be established against the complete full-text prior-art set.

2 Evidence corpus and study-design status

2.1 Search provenance and selection status

The frozen 31 October 2026 source archive contains 21 native Scopus, Web of Science and IEEE Xplore exports comprising 8,326 provider record rows. A conservative identity pass groups these into 5,440 *provisional* bibliographic work identities, preserving unresolved DOI/title collisions, conference-to-journal relationships, corrections and multi-record aliases. The identity-group count is not yet a fully adjudicated deduplicated-study denominator. A previous priority frame of 599 identities was used to organize deeper source retrieval, not to define the full included population.

As of the independent 7 November R47 Batch27 source-and-custody readback, the two nonoverlapping AI-assisted source-screen workloads have completed 2,588/2,588 and 1,375/2,588 records, respectively (3,963 of 5,176 assigned *screening records*). Those 5,176 assignments are worker source-screen workload rows, not 5,176 unique eligible clinical-style studies, and should not be substituted into a PRISMA flow. No completed full-text eligibility round, independent second-human screener, agreement statistic or final included-study count is claimed. PRISMA 2020 and the PRISMA-S search-reporting extension are planned reporting targets [1, 2]; the latter requires a reproducible database/platform/query/date record, beyond an assertion that a search was performed.

This version concentrates on a bounded reproducible science demonstration rather than extrapolating the partial screen to prevalence estimates. The executed illustrative source set is purposive. Three reports are matched to frozen identities and one original-publisher report, Çalışan (2026), was discovered *after the searches were frozen* despite its publisher publication date of 21 October 2026. It is a separately labeled contemporary source, *not silently inserted* into the 31 October export frame. An exact DOI/title lookup against the archived 5,440-identity metadata register returned no match, which establishes that narrow register fact but does not establish why a record was absent from the original institutional query results.

2.2 Experiment-indexed evidence record

Each proposed record has a stable source-to-result path:

Report and version ⟶ source release / annotation origin ⟶ physical acquisition parent ⟶ image / crop / augmentation descendants ⟶ train–validation–test assignment ⟶ model / checkpoint-selection exposure ⟶ metric, denominator and bounded claim

A record separates *reported* facts from *replayed* facts. In particular, author statements of module-grouped sampling are meaningful evidence and are recorded positively; they are not rebranded as independent raw-manifest verification. Conversely, unreported image-parent mapping is an identifiability limit, *not proof* of leakage or misrepresentation. The record retains a source URL, source tier, exact source section/table, and missing-value state for every stronger inference.

Table 1: Evidence tiers and defensible claims. Passing a lower-tier source check never automatically grants a stronger validation certificate.

Evidence tier	What it may establish	What it cannot alone establish
Provider or publisher abstract	Report identity, modality, indicative task, author-quoted headline outcome	Detailed split manifest, physical-parent independence, complete test denominator
Original Methods and tables	Author-described source population, split policy, checkpoint selection, task and metric values	Independent correctness of unprovided image-to-module ancestry
Owner code and versioned metadata	Release/commit identity and inspectable processing, when corresponding artifacts exist	Same experimental sample identity across papers without an explicit join
Raw image–parent replay	Literal zero cross-partition overlap for a defined mapped parent and version	Universal site transfer or absence of every possible nuisance/selection dependence

2.3 A source-only, reproducible diagnostic

To demonstrate why the checkpoint rule belongs in every result record, we re-tabulated Table 5 of Bommes et al. [14]. Its nine source-to-target *cross-plant* rows compare a contrastive k-nearest-neighbor method with a cross-entropy classifier. For each task and classifier the table reports three AUROC percentages: the model selected using validation plant 0, model selected using validation plant 1, and the *Best* score selected on labeled target data. This yields $9 \times 2 \times 3 = 54$ printed metric cells. The authors explicitly distinguish the target-best choice from feasible selection on another labeled PV plant; their published design also reports module-level train/test grouping and cross-plant evaluation. We do not retrain their models or re-estimate confidence intervals from per-run data.

For source–target task t, classifier $m \in \{C, E\}$ and selection condition $s \in \{0, 1, T\}$, let U_{tms} be the *printed* AUROC percentage. Define the within-condition method separation and a descriptive selection-condition difference as

$$\Delta_{ts} = U_{tCs} - U_{tEs}, \tag{1}$$

$$G_{tms} = U_{tmT} - U_{tms}, \quad s \in \{0, 1\}. \tag{2}$$

Here T is the target-label-selected Best, not a blind test estimator. G is *not* an experimental causal effect or a corrected estimate of generalization; it compares different reported checkpoint choices. A strict reversal is counted only when the sign of Δ_{tT} differs from the signs of both Δ_{t0} and Δ_{t1}. The printed two-decimal precision is retained. The supplied reproduction script reads the hash-pinned 54-cell extraction and checks the task count, sign flips and extrema.

3 Source-linked methodological results

3.1 Checkpoint choice changes the descriptive ordering

Figure 1 compares the two classifiers at each reported selection condition. It plots *within the same source–target task* rather than aggregating accuracies across heterogeneous studies. The signed method separation reverses between target-best and both validation-plant selectors on B→E, B→F and F→A: **three of the nine published cross-plant tasks**. This is an illustrative property of that study's printed results, not an estimate of how often such reversals occur in PV research. In B→E the target-best difference is only -0.02 AUROC percentage points; its small magnitude is important when interpreting the apparent winner. The corresponding B→F and F→A target-best separations are $+0.50$ and -1.93 points, respectively.

The largest reported target-best minus validation-plant differences are **11.71** points for contrastive and **19.73** points for cross-entropy, both on F→A relative to validation condition 1 (73.29 versus 61.58 and 75.22 versus 55.49, respectively). These values come from simple subtraction of Table 5 entries and do not establish target leakage, incorrect methodology, or an inflation rate. Indeed, the authors explicitly discuss why knowing the target's labels is unavailable in deployment and investigate validation on another plant. A careful review should use this source as a *positive disclosure case*: when selection conventions are transparent, the scope of the reported score is identifiable. A distinct interval check that permits each two-decimal reported AUROC cell to vary by up to 0.005 percentage points (and a pairwise method difference by 0.010) retains all three

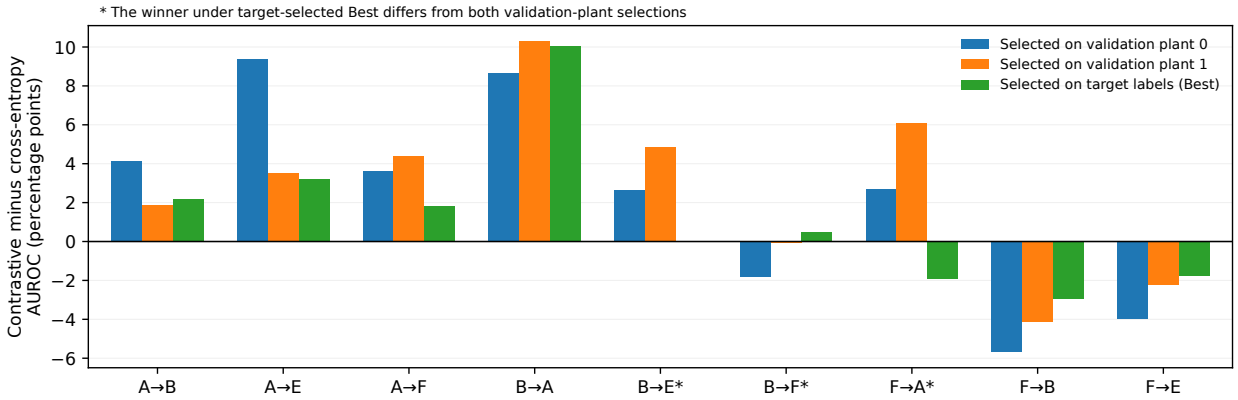


Figure 1: Difference in printed AUROC between contrastive and cross-entropy classifiers for nine source–target PV plant tasks under three different published checkpoint-selection conditions. Positive values favor contrastive; negative values favor cross-entropy. Asterisks identify strict direction changes versus *both* validation-plant conditions. The original authors report these outcomes; our contribution is the source-bound, transparent separation by checkpoint rule, not a new PV experiment.

printed ordering reversals. This is *display-rounding robustness only*, not a bootstrap, uncertainty of the underlying three runs, or statistical significance.

3.2 Image grouping, physical grouping and task support are distinct

A second original-source case reports 1,560 thermal images and a 70/15/15 *module-wise* partition for a six-class classification exercise; the same paper describes a separate 124-image hotspot-segmentation reference subset [15]. Its reported Random Forest accuracy of 97.6% is a classification result, not a segmentation result or a newly tested camera/site performance guarantee. The published grouping policy is informative, but an independent complete mapping of images to modules and new facilities was not replayed here.

Bayat Toksoz et al. report five-fold group-aware EL classification across 17,377 image observations and 17,324 available source-group identifiers [16]. The arithmetic difference, 53, does *not* identify 53 duplicates, 17,324 independent modules, or physical-parent disjointness. The same publication reports macro-F1 of 79.92% for one parameter-efficient configuration and 78.26% for full fine-tuning within its defined grouped-CV setting; the two training-memory figures 1,727 MB and 2,478 MB were obtained with different batch sizes, so their difference is not an isolated quantization effect. These are useful reproducible source observations and specifically bounded comparisons.

A separately tracked contemporary paper reports 0.8876 internal AUROC for a PVEL-AD held-out test cohort, followed by a transferred calibrated anomaly score saturating to one on all 2,624 ELPV images; its primary endpoint-label subset has 2,223 cases, for which the author reports AUROC 0.5000 [19]. This directly illustrates why a held-out score on one dataset cannot be silently presented as an externally valid score on a second dataset. The source itself distinguishes its subsequent uncalibrated post-hoc diagnostics from the original zero-shot calibrated transfer. The paper was discovered after the database-search freeze although the publisher dates its publication to 21 October 2026, ten days before that freeze. An independent literal DOI and title check of the original 5,440 provisional identities found neither. This is a bounded *search-frame coverage observation*, not proof that the October search strategy failed or that providers indexed the article by the query date. Its results remain contextual only until the search/additional-source and eligibility record are reconciled.

3.3 Source-version alignment is a separate identifiability gate

An illustrative 2026 preprint's public abstract describes a four-class, two-modality SolarFCD dataset comprising 4,435 images [17]. Its linked official code repository README, as inspected at its publicly available version, describes 3,742 images and a train/validation/test layout of 3,030/337/375, which correctly sums to 3,742 [18]. The two public descriptions differ by 693 images. Without the precise full paper protocol, archival dataset version and image-ID correspondence, this is a *source-release alignment question*, not a demonstration of a data error, leakage, or invalid reported model accuracy. The example also shows why an author's current

Table 2: Illustrative source-to-claim distinctions. Entries are individual experimental contexts, not a random or complete study sample.

Source	Reported experiment	Defensible source statement	Stronger claim remaining unresolved
Bommes et al. [14]	Cross-plant IR anomaly classification; two checkpoint regimes	Printed AUROC depends on checkpoint choice; author reports grouped modules and cross-plant tasks	Fully untouched target-label selection for target-best; independent raw parent and checkpoint replay
De Carlo et al. [15]	Thermal fault classification and separate mask subset	Authors describe module-wise splitting; distinct task denominators	New camera/site transfer; independently verified image–module map
Bayat Toksoz et al. [16]	Five-fold group-aware EL classification	Author-reported fold identifiers and metric family	Complete physical module ancestry and external-site evaluation
Calisan [19]	Internal PVEL-AD test; ELPV score transport	Source reports a failed calibrated transfer on a different EL family	An externally verified new-site failure rate, extrapolated population prevalence

Table 3: Two original-document reporting observations and the stronger result claims still unresolved. Neither observation proves defective study results or actual data leakage.

Report	Original source-visible statement	Claim not established
El-Rashidy (2022)	2,624 source images in Introduction; 2,426 in Experiments, with 44 panels	Actual image inclusion rule, source-to-fold physical parent map
Oulefki et al. (2024)	IoU printed as TP/(TP+FN), identical to recall	Whether the printed expression was used by the source code; corrected test-mask metrics

README should not be treated as proof of the images or source preprocessing used in a reported earlier experiment. SolarFCD is a contextual repository case; no new frozen-source inclusion is asserted.

3.4 Printed dataset and metric denominators are independent source gates

Two further original-document checks show why printed counts and measure definitions must remain attached to individual results. El-Rashidy's EL classifier paper describes a benchmark of 2,624 cell images in its Introduction but refers to 2,426 in the Experimental Results section; the latter describes 44 source panels and fivefold cross-validation [12]. The visible difference is 198 images, yet neither an image-exclusion crosswalk nor a fold-to-physical-panel manifest was obtained. This textual discrepancy is *not* proof of an incorrect result or a contaminated image split.

In a different thermographic segmentation paper, Oulefki et al. print the Jaccard/IoU formula as TP/(TP+FN) in Table 4, identically to the table's recall formula [13]. The conventional binary-mask IoU is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$. For a *synthetic* example with $\mathrm{TP}=20$, $\mathrm{FP}=10$, and $\mathrm{FN}=5$, the former evaluates to 0.800 and the latter to 0.571. The discrepancy concerns the *printed formula*, not independently inspected implementation: code and true masks were not replayed, and the original paper's Table 5 scores are neither recalculated nor invalidated here.

3.5 A reusable claim-support object, not a quality score

The accompanying JSON Schema and executable examples require the source and experiment identifiers, image modality, target task, evaluation unit, split-disjointness unit, source tier, physical-parent crosswalk status, target-label selection exposure, metric unit/support and requested claim. A rule refuses to upgrade a paper's author-described module-wise split to a RAW-REPLAYED physical-parent certificate unless a raw parent manifest is actually replayed and the claimed partition has verified zero overlap. A separate rule blocks an "untouched target-model selection" assertion when target labels are recorded as selecting the Best

checkpoint. These are transparent reporting rules and test fixtures, not empirical estimates of publication validity or a novel prediction model. No illustrative empirical case is given a raw-replayed physical-disjointness certificate in this draft.

4 Scientific interpretation and limitations

The central observation is a distinction between *observable reporting support* and *true but unavailable experimental conditions*. Physical independence is conditional on a parent relation (image, original capture, cell, module, batch, flight, date or plant) and on an explicit boundary between data used in fitting, tuning and outcome estimation. An author may correctly use a grouped split without publishing every manifest needed for independent verification. Conversely, the presence of multiple plants or many augmented images cannot by itself prove an untouched test cohort. The intent of the proposed record is to make these statements precise, not to discredit high-performing models.

This manuscript is not yet a completed systematic review. The initial W4 source screen remains incomplete; later full-text eligibility, source-depth evidence reconciliation, bibliographic identity arbitration and formal PRISMA/PRISMA-S reporting are unclosed. The early 599-group retrieval priority is *not* a representative sample from which to estimate the prevalence of leakage or validation practice. AI-assisted screening is reported as a single AI-assisted source-screen process; no second independent human reviewer or kappa is invented. Neither the currently available source texts nor the hash-matched bibliographic files demonstrate every trained checkpoint's physical disjointness. The eight published result contexts in the development pilot are purposively selected; 54 Bommes cells belong to *one* source paper and cannot be treated as 54 independent studies. Statistical significance, causal bias adjustment, pooled accuracy and site-transfer rates are not estimated.

The novelty claim has a further constraint: prior reviews already tabulate original and augmented PV image counts, split allocations, datasets and model results; several 2026 reviews also identify dataset heterogeneity and evaluation-metric comparability problems [3, 4, 5, 6, 7]. Prior-art comparison here is verified at heterogeneous source tiers (some publisher abstracts, one complete open full text) and is not an exhaustive claim that no predecessor has ever proposed a structured validation-audit checklist. A stronger final submission would systematically distinguish which competing works actually execute source-version/physical-parent/checkpoint/result-level checks, rather than saying only that they "mention" datasets.

5 Conclusion and next evidence gate

A citable methodological contribution is possible if the final paper demonstrates a consistent, reproducible mapping from published PV defect-image results to the specific experiment and statistical unit their evidence supports. The present bounded reproduction shows why checkpoint selection matters: in a transparent nine-task cross-plant study, the reported model ordering can depend on whether labels from another PV plant or the target plant determine the reported checkpoint. Complementary examples show that author-reported module-grouping, source-grouped folds and calibration transfer each address a different evidentiary question. The next necessary work is to complete the outstanding source and full-text screening and close the final PRISMA-S and identity gates while preserving exact source provenance and a publicly releasable version of the audit scripts. Strong claims about actual leakage, external generalization prevalence, or a pooled model leaderboard are not warranted by the present evidence.

Data, code and integrity

The R46 calculation baseline and R47 append-only audit supplements contain the frozen Pilot02 extracted-source ZIP and independent reconstruction scripts, the machine-readable nine-task selection table, source-bounded claim records, negative/positive rule fixtures and a SHA-256 manifest. The 21 institutional database exports, publisher PDFs and private source abstracts are *not* redistributed in the journal-facing subset. A final license-appropriate repository deposit, complete study-selection flow, final author metadata and selected journal format are pending; the supplied controlled source is currently for internal research and author review only.

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